WHAT HIDES MUSICAL GEOMETRY IN AUDIO REPRESENTATIONS?

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ABSTRACT

Probing studies establish that audio representations encode musical attributes such as key, and this is routinely read as evidence that the representations are musically organised. We show these are different claims. Measuring the same 24 musical keys on the same corpus three ways, by linear decodability, by the geometry of class centroids, and by the geometry of a linear readout's weights, we find they dissociate. A log-CQT decodes key better than MERT-v1-330M, at .492 against .456, while its centroid geometry shows no circle-of-fifths structure at all, yet the same structure is strongly present in its probe weights. We then identify two distinct mechanisms that hide musical structure from the ambient metric. Sample-limited masking is class-independent nuisance variance. It averages away, and it alone accounts for clip-level key geometry, which moves by a factor of 13.4 with centroid sample size but only 1.7 with a thousandfold longer pooling window. Metric masking is nuisance that is a function of the class. It does not average away at any sample size, and a rank-4 projection fitted on GiantSteps keys, never on notes, recovers octave equivalence in individual NSynth recordings using under 0.3% of the variance. The distinction is not specific to music representations.

***Index Terms*—** music representation learning, probing, representation geometry, key detection, octave equivalence

1. INTRODUCTION

Self-supervised audio encoders support strong linear probes [1] for key, pitch, genre and instrument [2, 3], and the usual inference is that the attribute is *organised* in the representation. Music makes that inference unusually testable, because music theory specifies not just which classes exist but how they are related. Keys a perfect fifth apart share six of seven scale degrees and sit adjacent on the circle of fifths [4, 5], and pitches an octave apart are the same pitch class [6]. A representation can therefore encode key perfectly while placing C and G no closer together than C and F$\sharp$.

We ask when that theoretical structure is visible in the geometry a downstream model actually consumes, and find that decodability, ambient geometry and readout geometry come apart (Sec. 4.1). We then separate two reasons a representation can contain musical structure that its metric does not show. Write an item of class y as

$$z = s_y + n_y + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid y] = 0, \tag{1}$$

with s_y the structure of interest, ε nuisance independent of the class, and n_y nuisance that is a *function* of the class. The two nuisance terms behave differently under the two operations available to an analyst. Averaging many examples of a class drives ε towards zero, so structure hidden by ε appears once enough examples are pooled; averaging never removes n_y, which is present in every example of the class by construction. A linear projection chosen to suppress n_y does the reverse. We show that clip-level key geometry is entirely the first case (Sec. 4.2) and note-level octave equivalence entirely the second (Sec. 4.3).

Our contributions are a three-way measurement that separates musical information, ambient geometry and readout geometry with synthetic controls that make a null interpretable, a cross-dataset projection that exposes octave equivalence no amount of averaging recovers, and the negative result that this structure is redundantly distributed rather than confined to a compact subspace.

2. RELATED WORK

Linear probes are the standard instrument for asking what a representation encodes [1], and music audio encoders are routinely evaluated this way, with MERT [2] and the MARBLE benchmark [3] reporting probe accuracy for key, pitch, genre and instrument. Hewitt and Liang [7] show that a probe's success has to be read against a control task holding accuracy fixed while removing the structure of interest, and we adopt that design for representation geometry rather than for probe complexity.

The relational structure we test against is the one music cognition established for human listeners, in which perceived key relations follow the circle of fifths and Krumhansl–Kessler profiles [4, 5] and pitch is organised on a helix that separates height from chroma [6]. Chroma features [8, 9] build octave equivalence into the input by construction, which makes

Code, pre-registrations and all run artefacts: `https://github.com/Junst/topo-music`. Every threshold and kill criterion reported here was registered before the corresponding run.

them a positive control rather than a competitor, while tonality-aware objectives such as STONE [10] impose key equivariance during training, whereas the question here is what a representation trained without such a constraint already contains. We also include the codec that music generation systems consume [11, 12], since the geometry a generator sees is that latent's.

3. MEASUREMENTS

Representations. MERT-v1-330M [2] layers 4 to 24, EnCodec 32 kHz [11], the codec MusicGen consumes [12], and a log-CQT [8, 9] as an input floor. A chromagram serves as the music-theoretic positive control. Clips are pooled to a single vector by a mean over frames.

Key-centroid geometry. On GiantSteps [13], with 7035 clips covering all 24 keys, we average clip embeddings within key to obtain 24 centroids and rank-correlate their pairwise Euclidean distances against three reference structures: circle-of-fifths distance, the Krumhansl–Kessler key-profile distance, and semitone distance. Significance uses a 2000-draw permutation null over key labels, and a partial Spearman additionally removes mel-spectral centroid distance. Reporting the semitone reference alongside the fifths reference is essential, because a generic tendency for nearby keys to look alike would raise both.

Readout geometry. The same probe, a multinomial logistic regression under 5-fold cross-validation, yields 24 class weight vectors, and we apply the identical measurement to their pairwise cosine distances. This asks whether a linear readout can reach structure the ambient metric does not express.

Octave equivalence. On NSynth [14] we use real recordings only. Pitch shifting by signal processing would build the answer into the stimulus. For each instrument we fix an anchor set of pitches supporting every transposition up to 25 semitones, so the anchor set is constant across transpositions, and measure the mean cosine similarity between a note and its transposition by k semitones. Three statistics are reported together, as registered in advance: a local octave contrast, comparing the similarity at 12 semitones against the mean of 11 and 13; a strict contrast, comparing it against the largest similarity anywhere from 8 to 11 semitones; and the coefficient on a twelve-semitone periodic term in a fitted curve. Confidence intervals are bootstrapped over *instruments*. The local contrast alone is not sufficient, since a positive periodic coefficient with no local peak at 12 semitones is a curve-fitting artefact, so we require the three to agree in sign.

Subspaces. Each subspace is an orthonormal basis of the top d shrinkage-LDA discriminant directions for a stated target, with d at most 11, since any linear map to a twelve-class target has rank at most 11. We report the fraction of total variance the subspace holds and, as a control, matched random orthonormal subspaces, because projection changes the metric whatever the directions are.

Table 1. Key information, ambient geometry and readout geometry come apart. Same 7035 GiantSteps clips, same 24 keys. *acc* is 24-way key accuracy (chance .042); ρ_5^{cent} is the circle-of-fifths correlation among the 24 key centroids, i.e. the ambient metric; ρ_5^W is the same correlation among the probe's 24 class weight vectors, i.e. what a linear readout reaches. The controls fix the scale for reading the two geometry columns.

representation	acc	ρ_5^{cent}	ρ_5^W
Controls			
analytic fifths (metric ctrl)	0.274	+0.977	+0.986
random features (no info)	0.055	+0.009	+0.033
orthogonal key codes (no tonal rel.)	0.651	−0.008	−0.005
Spectral front ends			
log-CQT, 84 bin, dB	0.492	−0.035	+0.560
log-CQT, 84 bin, per-frame norm	0.486	−0.117	+0.671
octave-summed, 12 bin, dB	0.407	−0.106	+0.305
chromagram (fold + norm)	0.451	+0.475	+0.692
Neural codec			
EnCodec 32 kHz	0.404	+0.177	+0.648
MERT-v1-330M			
layer 4	0.451	+0.445	+0.412
layer 12	0.456	+0.468	+0.428
layer 16	0.440	+0.395	+0.409
layer 24	0.418	+0.294	+0.484

4. RESULTS

4.1. Information, metric and readout dissociate

Table 1 measures all three on the same clips and keys, and the control block is what sets the scale for reading it. An analytic circle of fifths scores close to 1 in every column, so a null elsewhere is a null of the representation and not of the method, while random features carry no information and show nothing anywhere. The load-bearing control is the third one, in which orthogonal per-key codes decode key better than any real representation, at .651, with classes equidistant by construction, and yet their weight geometry is -0.005. In the sense of [7] this is a control task: it holds accuracy fixed and removes the structure, so what remains in the column is attributable to the representation. It rules out the natural sceptical reading, that keys a fifth apart are confusable whatever was fitted, so any accurate 24-way classifier ends up with fifths-structured weights.

The dissociation that follows is unambiguous, in that the log-CQT has the highest key accuracy of any real arm and no fifths structure in its centroids, at -0.035 with p of .52, while its probe weights carry that structure strongly, at 0.560 with a z of 9.1. Tonal relations are therefore linearly accessible in a log-spectrogram and simply absent from its metric, which is a phenomenon that requires no learned encoder at all.

What MERT does is move structure between columns rather than create it. Its readout geometry, between 0.41 and 0.48, is *lower* than that of the log-CQT, EnCodec or the chromagram, while its centroid geometry is far higher. Across depth, tonal relational geometry weakens, from 0.445 at layer

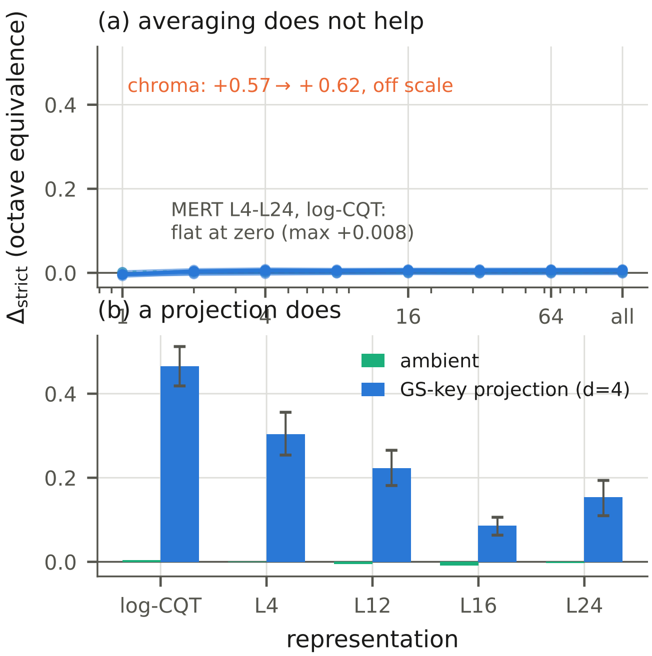

Fig. 1. Averaging removes sample noise; projection removes metric anisotropy. Top: averaging notes of the same pitch into a pitch centroid does not produce octave equivalence at any sample size, with every learned and spectral arm saturating by four notes at approximately zero and MERT L24 staying negative. The chromagram, off scale, has octave equivalence already in single notes. Bottom: a rank-4 projection fitted only on GiantSteps track tonics, with no notes and no NSynth, applied unchanged to individual NSynth recordings. Both panels share a vertical axis, since rescaling the top one would invent structure that is not there. Error bars are instrument-bootstrap 95% intervals.

4 to 0.294 at layer 24, while major and minor become more separable, from 0.216 to 0.328. We report this as a correlation rather than a mechanism, since key accuracy does not improve along the way, running .451, .456, .440, .418, and the mode effect is -0.054 and not significant in the chromagram control, so the growing mode axis is not pitch-class based and is at least as consistent with production and instrumentation in an all-EDM corpus.

4.2. Sample-limited masking

The clip-level result invites a temporal reading, in which keys are properties of seconds of music and pooling more audio should therefore reveal them, but that reading does not survive the measurement. Fig. 2(a) sweeps the pooling window from one frame to the full 20 s clip, drawing one window per clip at every length so the number of clips behind each centroid is held constant. A single 13 ms MERT frame already yields a correlation of 0.276, which is 59% of the 20 s value, and the chromagram is flat to within 3%. There is no knee and no threshold.

The clip-level measurement averages twice, and only one of those averages was being varied. Fig. 2(b) holds the window at the full clip and sweeps the number of clips per centroid, giving 0.035 at one clip and 0.468 at all of them for MERT L12, a factor of 13.4 against 1.7 for a thousandfold longer window. At one clip per key, which is the condition the pairwise note-level test is in, the geometry is indistinguishable from zero for every arm.

This is Eq. (1) with ε dominant, because what differs between two GiantSteps tracks in the same key, namely production, instrumentation and subgenre, is largely unrelated to the key and cancels in the mean, so what was missing at one clip per key was samples rather than audio.

4.3. Metric masking

If low between-item signal-to-noise were the whole story, the same averaging should expose octave equivalence on NSynth, and we pre-registered that prediction in advance of the run that refutes it. Fig. 1 sweeps notes per pitch centroid, and every arm saturates by four notes at approximately zero, MERT L24 remains negative, and the chromagram needs no averaging at all, at 0.566 from single notes. Register and spectral envelope are functions of pitch, so they survive into every pitch centroid, so averaging over instruments cancels the instrument but never the octave, which is exactly the behaviour of the n_y term.

What removes n_y is a projection, and the evidence is cross-dataset and cross-task. We fit LDA directions on GiantSteps track tonics only, with no notes, no NSynth and no octave information, and apply them unchanged to individual NSynth recordings with no averaging. At rank 4 the strict octave contrast moves from -0.005 to 0.223 for MERT L12, from -0.001 to 0.304 for layer 4, and from 0.004 to 0.466 for the log-CQT, with the other two statistics agreeing in sign and bootstrap intervals excluding zero. Matched random subspaces stay between -0.019 and 0.007. The subspace uses between 0.11 and 0.24% of MERT's variance and 1.85% of the log-CQT's.

It would overstate the case to say key supervision *forces* pitch-height invariance, since GiantSteps key correlates with subgenre, tempo and production, and an LDA may use any of it. The transfer is what licenses the claim, and it licenses a stronger one: structure that survives the move to an independently recorded isolated-note corpus cannot be solely a GiantSteps-specific nuisance correlation. The invariance is demonstrated across datasets rather than guaranteed by the objective. Objectives that build key equivariance in by construction [10] are a complementary route we do not pursue here.

A registered prediction that failed. We predicted that removing the subspace from the representation would at least halve clip-level fifths geometry. Against a baseline recomputed on

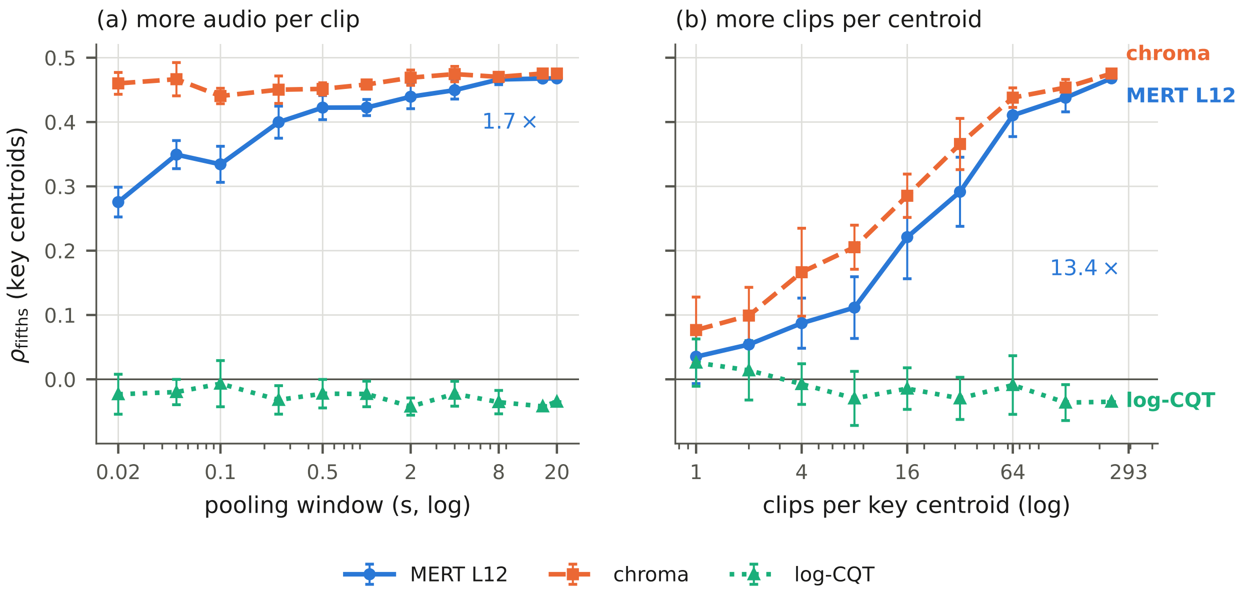

Fig. 2. Clip-level key geometry is set by centroid sample size, not by temporal context. Both panels plot the same quantity on the same vertical axis. (a) A thousandfold longer pooling window moves MERT L12 by a factor of 1.7 and moves the chromagram by 3%, and a single 13 ms frame already carries 59% of the 20 s value. (b) Increasing the number of clips averaged into each key centroid moves it by a factor of 13.4, and at one clip per key, the condition a pairwise note-level test is in, the geometry is absent. Error bars are one standard deviation over 5 draws in (a) and 10 in (b).

the same held-out clips it does not move it, going from 0.127 to 0.119 for MERT L12 and from 0.118 to 0.114 for layer 4, and only the twelve-dimensional chromagram collapses, which is a dimensionality effect. Tonal structure is therefore low-energy and invisible in the ambient metric, but it is *not* confined to a compact subspace. Many overlapping direction sets carry it, and a low-dimensional projection denoises a redundantly distributed signal rather than isolating a unique one.

5. DISCUSSION

Three properties the probing literature often treats as one, namely information availability, metric visibility and readout accessibility, are separable, and separating them changes what a probe result licenses: a high probe accuracy says the structure is linearly recoverable, and says nothing about whether it governs the distances a model relying on similarity or retrieval consumes.

The two masking modes suggest different remedies. Sample-limited masking is a property of the evaluation, and the fix is more examples per class, not longer audio. Reporting the number of items behind each centroid should be standard, since the same encoder scores 0.035 or 0.468 depending only on that. Metric masking is a property of the representation itself, which no evaluation protocol can remove and only a learned transform addresses.

Nothing in Eq. (1) is specific to music, since semantic relations in vision may equally be linearly available while Euclidean geometry is dominated by pose or background, and phonetic structure in speech may be masked by speaker and channel. The distinction between information availability and metric visibility plausibly extends beyond music representations.

Limitations. GiantSteps is entirely electronic dance music, where key correlates with subgenre. The cross-dataset transfer in Sec. 4.3 constrains that confound but does not remove it, and a key-balanced, genre-diverse corpus would settle it. NSynth is monophonic and either synthesised or sampled from isolated instruments, so its nuisance structure is unusually clean. Our subspaces are linear by design, and a nonlinear transform might expose more, at the cost of the interpretability that makes the class-coupled nuisance account testable. Finally, the mode effect is reported but not explained, and we do not claim it is harmonic.

6. CONCLUSION

Musical structure can be present in a representation, linearly accessible and still invisible in its geometry, for two reasons calling for two different operations. Averaging over examples removes class-independent nuisance and accounts entirely for clip-level key geometry, while a rank-4 projection fitted on another dataset and task removes class-coupled nuisance and recovers octave equivalence that averaging never reaches. Neither operation creates the structure it exposes, and knowing which of the two a null calls for is what separating them buys.

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